



Computing the convex envelope of bivariate piecewise linear-quadratic functions in linear time

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Abstract

We compute the convex envelope of (nonconvex) bivariate piecewise linear-quadratic (PLQ) functions (bivariate quadratic functions defined on a polyhedral subdivision). Our algorithm is composed of the following steps: (1) compute the convex envelope of each quadratic piece obtaining piecewise rational functions (quadratic divided by linear function) defined over a polyhedral subdivision; (2) compute the (Legendre-Fenchel) conjugate of each resulting piece to obtain piecewise quadratic functions defined over a parabolic subdivision; (3) compute the maximum of all those functions to obtain the conjugate of the original PLQ function as a piecewise quadratic function defined on a parabolic subdivision; (4) compute the conjugate of each resulting piece; and finally (5) compute the maximum over all those functions to obtain the convex envelope (biconjugate) as rational functions (quadratic divided by linear function) defined over a polyhedral subdivision. Our contribution includes a practical algorithm running in linear time, and proving that the convex envelope is a piecewise rational function.

Keywords Global optimization · Conjugate · Convex envelope · Piecewise quadratic function

1 Introduction

There are several motivations for the present work. One goal is to obtain tight lower bounds for relaxation in global optimization. The convex envelope, which is equal to the biconjugate, provides the tightest relaxation when one wishes to solve a global optimization problem. The convex envelope also provides information on the set of minimizers. If we denote by $\operatorname{argmin} J$ the set of $x \in X$ minimizing J on X (possibly,

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the empty set), we easily see that

$$\overline{\text{conv}}(\text{argmin } J) \subset \text{argmin } \overline{\text{conv}}(J),$$

with $\overline{\text{conv}}(J)$ denoting the closed convex envelope of J , with equality for example when $\overline{\text{conv}}(J)$ is 0-coercive [1, Theorem 6]. A large literature is available on global optimization and the search for tight relaxation; we refer to [2] (especially Section 4.2 on convex envelopes) for a detailed introduction and more details on relaxations relevant for our context.

Another goal is to understand the structure of the biconjugate of piecewise linear-quadratic (PLQ) functions (bivariate functions defined on a union of polyhedral set on each of which the restriction of the function is quadratic). Such understanding is of interest in itself.

Computing the convex envelope of a function is a hard problem; even computing the convex envelope of a multilinear function over a unit hypercube is NP-Hard [3]. However, results for specific functions exist in the literature, in particular for quadratic bivariate polynomials [4–8], and for convex envelopes of bilinear functions over triangles, rectangles and special polytopes [9–13]. It is an active subject of research [14].

PLQ functions play a significant role in variational analysis [15, Section 10E, Section 11D, p. 440] due to the availability of calculus rules [15, Example 11.28, Proposition 11.32, Corollary 11.33, Proposition 12.30], and due to their duality property [15, Theorem 11.42, Example 11.43, Theorem 11.42]. Compared to piecewise linear functions, PLQ functions capture the curve of the original function more accurately. The set of convex PLQ functions is closed under common convex operators, in particular under the Legendre-Fenchel transform; see [15, Page 484], [16, Proposition 3, p. 17] and [17].

The early idea of the computation of convex transforms can be traced back to [18]. However, development of most of the algorithms in computational convex analysis began with the computation of the conjugate with the Fast Legendre Transform [19], which is studied in [20, 21]. A linear-time algorithm is introduced in [22]. Those algorithms handle nonconvex functions but are restricted to grid domains. Extension to nongrid domains for convex functions only are proposed in [17, 23] with a generalization to the partial conjugate in [17]. Computation of the conjugate of convex univariate PLQ functions has been well studied and linear time algorithms have been developed, both for the PLQ [24] and the graph-matrix (GPH) data structures [25]. All in all, we are not aware of algorithm to compute the conjugate of nonconvex PLQ functions over nongrid domains.

Implementation of conjugate computation algorithms can be found in the CCA library [26, 27] that contains efficient algorithms to manipulate convex functions and to compute fundamental convex analysis transforms arising in the field of convex analysis. Algorithms to compute the conjugate numerically on grids have been based on either parameterization [28], manipulation of graphs (GPH model) [17], or the computation of the Moreau envelope [29]. More complex operators such as the proximal average operator [30, 31] can be built by using a combination of addition, scalar multiplication, and conjugacy operations.

A numerical library to determine in linear time whether a PLQ function is convex [32] is available in MATLAB [33]. This library is the first to handle general piecewise quadratic functions that are not necessarily continuous and are defined on any polyhedral subdivision without approximating them with a grid.

Computational Convex Analysis has many applications in the fields of image processing, network communication, PDE, geographic information systems, computer aided design, molecular biology, medical imaging, computer graphics and robotics; see the survey [34].

We propose the first algorithm to compute the convex envelope of a general bivariate PLQ function not necessarily convex without approximating it with a grid. We assume the pieces are triangular; this assumption is not restrictive since we can always triangulate convex polyhedral sets in linear time [35, p. 49]. Our algorithm is split in four steps. In Step 1 we compute the convex envelope of each piece of a PLQ function f using formulae derived in Appendix A using the method given in [6]; in Step 2, we apply [36] to compute the conjugate of each resulting piece; and in Step 3, we compute the maximum of all the resulting piecewise functions to obtain the conjugate f^* [37, 38]. Step 4 computes the biconjugate to obtain the convex envelope by first computing the conjugate of each piece and then computing the maximum of all the resulting piecewise functions.

Note that one cannot simply apply [38] twice since applying it once does not result in a PLQ function, but in a function defined on a parabolic subdivision for which we are not aware of any algorithm.

Our contributions are as follow: (1) developing formulae to compute in linear time the convex envelope and conjugate of each piece (this steps improves Locatelli's method [6] that has a worst case exponential time), (2) proving the entire algorithm runs in linear time with respect to the number of pieces of the input PLQ function, (3) proving the convex envelope of a nonconvex PLQ function is a piecewise rational function (quadratic over linear) on a polyhedral subdivision (although the type of subdivision was known, the type of functions was not); and (4) releasing an open-source code that uses symbolic computation and rational arithmetic to avoid floating point errors.

We note that [6] focuses on existence rather than efficient computation while [7] provides an algorithm that solves the KKT conditions to obtain the convex envelope of some bivariate functions over polytopes which can be done in $O(n)$, where n is the number of entities. This gives a linear-time algorithm to compute the convex envelope of a bilinear function over a polytope while we are giving a linear time algorithm to compute the convex envelope of a PLQ function, *i.e.*, a piecewise quadratic function defined on a polyhedral partition.

We begin by giving some preliminaries in Sect. 2, describing the structure of the conjugate of a PLQ function in Sect. 3 and then the method to compute the domain and the conjugate expressions in Sects. 4 and 5. We present examples in Sect. 7, and conclude with future work in Sect. 8.

2 Preliminaries

The *domain* of a function $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ is the set $\{x : f(x) < +\infty\}$, and the function is called *proper* if it has nonempty domain. The *indicator function* of $C \subset \mathbb{R}^n$ is denoted $I_C : \mathbb{R}^n \rightarrow \{0, \infty\}$ with $I_C(x) = 0$ if $x \in C$ and $+\infty$ otherwise.

A function is called a *piecewise function* if it can be written $f(x) = f_i(x)$ for $x \in R_i$ with $\cup_i R_i = \mathbb{R}^n$. We call the pair (f_i, R_i) a *piece*. Recalling [15, Definition 10.20], a function $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ is called *Piecewise Linear-Quadratic (PLQ)* if there exists a finite number of polyhedral sets R_i with $\text{dom } f = \cup_i R_i$ and f restricted to R_i , noted f_i , can be written $f_i(x) = 1/2x^T Q_i x + q_i^T x + \kappa_i$ with $\kappa_i \in \mathbb{R}$, $q_i \in \mathbb{R}^n$, and $Q_i \in \mathbb{R}^{n \times n}$ a symmetric matrix. We use column vectors with $q^T x$ denoting the dot product of row vector q^T with column vector x . PLQ functions enjoy several properties, e.g., their domain is closed, they are lower semicontinuous, and continuous relative to their domain [15, Proposition 10.21].

An *underestimator* is any function that lies below a given function. The *closed convex envelope* is the largest lower semicontinuous convex underestimator of f .

We note

$$f^*(y) = \sup_{x \in \mathbb{R}^n} (x^T y - f(x))$$

the (*Legendre-Fenchel*) *conjugate* of f . The *biconjugate* is noted f^{**} , and we recall that when f is proper, closed and convex, $f = f^{**}$, otherwise f^{**} is the closed convex envelope of f [15, Theorem 11.1].

To describe the domain of piecewise functions, we note a *hyperplane* $H \subset \mathbb{R}^d$ a set of the form $H = \{x \in \mathbb{R}^d : \alpha^T x - \beta = 0\}$, and a *halfspace* $H \subset \mathbb{R}^d$ a set of the form $H = \{x \in \mathbb{R}^d : \alpha^T x - \beta \leq 0\}$ or $H = \{x \in \mathbb{R}^d : \alpha^T x - \beta \geq 0\}$; where the vector $\alpha \in \mathbb{R}^d \setminus \{0\}$ is called a *normal vector* to H .

A *polytope* is a convex combination of finite numbers of vertices $\{v_1, \dots, v_k\}$. We write $\text{conv } V = \{\lambda_1 v_1 + \dots + \lambda_k v_k : \lambda_i \geq 0, \sum_{i=1}^k \lambda_i = 1\}$ or in matrix notation $\text{conv } V = \{V\lambda : \lambda_i \geq 0, \sum_{i=1}^k \lambda_i = 1\}$, where V is the matrix formed by column vectors. Note that by our definition, polytopes are always bounded.

A *polyhedral cone* is defined as a conic combination of a finite numbers of directions $\{d_1, \dots, d_m\}$; we write $\text{cone } C = \{\mu_1 d_1 + \dots + \mu_m d_m : \mu_i \geq 0\}$, and $\text{cone } C = \{D\mu : \mu_i \geq 0\}$; where D is the matrix formed by column vectors d_i .

A *polyhedral set (polyhedron)* $P \subset \mathbb{R}^d$ is defined as the intersection of a finite number of closed halfspaces and hyperplanes; it has the H -representation

$$P = \{x \in \mathbb{R}^d : Ax \leq b\}$$

with $A \in \mathbb{R}^{m \times d}$ and $b \in \mathbb{R}^m$. We can also describe a polyhedral set in Vertex-representation. According to the Minkowski-Weyl Theorem [15, Corollary 3.53], a polyhedral set can be written as a sum of a polytope and a polyhedral cone. Therefore, in V -representation a polyhedral set P is described as

$$P = \text{conv } V + \text{cone } C = \left\{ V\lambda + D\mu : \lambda \in \mathbb{R}^k, \lambda_i \geq 0, \mu_i \geq 0, \sum_{i=1}^k \lambda_i = 1 \right\}.$$

Since the intersection of convex sets is convex, polyhedral sets are convex.

A d -dimensional face of a convex set $C \subset \mathbb{R}^n$ is a d -dimensional convex subset C' of C such that every (closed) line segment in C with a relative interior point in C' has both endpoints in C' [39, Page 162]. The empty set and C itself are faces of C . The two-dimensional faces of C are called faces. The one-dimensional faces of a convex set C are called edges. For $u, v \in \mathbb{R}^d$, $u \neq v$ an edge can be written as

$$\left\{ x \in \mathbb{R}^d : x = \delta_1 u + \delta_2 v, \delta_1 + \delta_2 = 1, (\delta_1, \delta_2) \in \Delta \subset \mathbb{R}^2 \right\}.$$

An edge in $\mathbb{R}^2$ is a segment if $\Delta = \mathbb{R}_+ \times \mathbb{R}_+$ where $\mathbb{R}_+ = \{\alpha \in \mathbb{R} : \alpha \geq 0\}$, a ray if $\Delta = \mathbb{R}_+ \times \mathbb{R}$, and a line if $\Delta = \mathbb{R}^2$. The zero-dimensional faces of a convex set C are called vertices and are the extreme points of C .

A convex set defined as the union of a finite number of polyhedral regions, namely $R = \bigcup_{i=1}^n R_i$, where $R \subseteq \mathbb{R}^2$, is said to be a polyhedral subdivision [40, Definition 1], if R_i is a polyhedral set and for any $j, k \in \{1, \dots, n\}$, $j \neq k$, $R_j \cap R_k$ is either empty, a vertex or an edge. (Polyhedral subdivisions eliminate degenerate subdivisions for which the intersection of two polyhedral regions is a strict subset of an edge, or of a face.)

Assume that $f : \mathbb{R}^2 \rightarrow \mathbb{R} \cup \{+\infty\}$ is a PLQ function and that $\text{dom } f = \bigcup_i P_i$ is a polyhedral subdivision. An entity is a d -dimensional face of P_i for some index i . An entity is either a vertex, an edge or a face of the polyhedral subdivision of $\text{dom } f$ [41].

We now define the geometric shapes required to describe the domain of the conjugate functions we obtain. A parabola $P_a \subset \mathbb{R}^2$ is a subset of the plane that can be written as

$$P_a = \{(x, y) \in \mathbb{R}^2 : ax^2 + bxy + cy^2 + dx + ey + f = 0\},$$

where $a, b, c, d, e, f \in \mathbb{R}$ are not all zero and satisfy $b^2 - 4ac = 0$. Note that our definition includes lines when $a = b = c = 0$, and the empty set when $a = b = c = d = e = 0$, $f \neq 0$; but excludes the entire plane since we impose that $(a, b, c, d, e, f) \neq 0$.

A parabolic region $P_a \subset \mathbb{R}^2$ is formed by the intersection of a finite number of parabolic inequalities. It can be written as

$$P_a = \{x \in \mathbb{R}^2 : a_i x^2 + b_i xy + c_i y^2 + d_i x + e_i y + f_i \leq 0, \forall i = 1, \dots, k\},$$

where $(a_i, b_i, c_i, d_i, e_i, f_i) \neq 0$, and $b_i^2 - 4a_i c_i = 0$. The following sets are special cases of parabolic regions: polyhedral sets, polyhedral cones, polytopes, edges, and vertices. The set $\{(x, y) \in \mathbb{R}^2 : y \geq x^2\}$ is an example of a non-polyhedral parabolic region while the set $\{(x, y) : y \leq x^2, y \geq 1\}$ is a nonconvex non-connected parabolic region.

A convex set $R \subset \mathbb{R}^2$ is called a parabolic subdivision if R can be written as the finite union of parabolic regions R_i , i.e., $R = \bigcup_{i=1}^n R_i$ and for any $j, k \in \{1, \dots, n\}$, $j \neq k$, the intersection $R_j \cap R_k$ is either empty or is contained in a parabola. A

polyhedral subdivision is a special case of a parabolic subdivision (since lines are a special case of parabolas).

A face of a parabolic region $P_a \subset \mathbb{R}^2$ is defined as the interior of a nonempty intersection of a finite number of parabolic inequalities. It can be written as

$$P_a = \{x \in \mathbb{R}^2 : a_i x^2 + b_i xy + c_i y^2 + d_i x + e_i y + f_i < 0, i = 1, \dots, k\},$$

where $(a_i, b_i, c_i, d_i, e_i, f_i) \neq 0$, and $b_i^2 - 4a_i c_i = 0$.

We define a parabolic function as a convex quadratic function with a zero discriminant. We write $q_p(x, y) = ax^2 + bxy + cy^2 + dx + ey + f \in \mathbb{R}^2$, where $a > 0$, $c > 0$, $b, d, e, f \in \mathbb{R}$ and $b^2 - 4ac = 0$.

Finally, we use standard definitions for subdifferentials, subgradients, and normal cones; see for example [39, 42–44].

While the convex envelope of a piecewise function is not always the same as the convex envelope of each piece, for bivariate quadratic polynomials we can leverage the convex envelope of each piece to obtain the convex envelope. Let us denote the i^{th} piece of a piecewise function f by (f_i, P_i) where f_i is a function and P_i is a set. From [44, Theorem 3.4.1], we have $(\inf_i f_i)^* = \sup_i f_i^*$. Using [44, Proposition 2.6.1] and the biconjugate theorem $\text{conv } f = f^{**}$, we get

$$\begin{aligned} \text{conv}[\inf_i (f_i + I_{P_i})] &= \text{conv}[\inf_i (\text{conv}(f_i + I_{P_i}))], \\ &= [\inf_i (\text{conv}(f_i + I_{P_i}))]^*, \\ &= [\sup_i ((\text{conv}(f_i + I_{P_i}))^*)]^*. \end{aligned}$$

Hence we can find the biconjugate of a bivariate PLQ function by the following steps:

1. Compute the convex envelope of each piece $\text{conv}(f_i + I_{P_i})$.
2. Compute the conjugate of each convex piece $[\text{conv}(f_i + I_{P_i})]^*$.
3. Compute the maximum of the conjugates over the entire PLQ function to obtain $f^* = \sup_i (\text{conv}(f_i + I_{P_i}))^*$.
4. Compute the biconjugate by computing the conjugate of the maximum of the conjugates over the entire PLQ function $f^{**} = (\sup_i (\text{conv}(f_i + I_{P_i}))^*)^*$.

Step 1 of this process, finding the convex envelope of each piece (f, P) with $f(x) = x^T A x + b^T x + c$, is computed as follow. We compute the eigenvalues of A . If they are both nonnegative, the function is convex and we are done. If they are both nonpositive, the function is concave and the convex envelope is obtained as the convex hull of the points $\{(x, f(x)) : x \text{ is a vertex of } P\}$. If A is indefinite (one positive and one negative eigenvalue), we note that $\text{conv } f(x) = \text{conv}(x^T A x) + b^T x + c$; see [43, p. 93]. Hence, we only focus on the quadratic part. We rotate the polyhedral set so that in the new basis, the function is now $(x_1, x_2) \mapsto x_1 \cdot x_2$.

We consider triangulations of the domain. When we restrict the bilinear function $(x_1, x_2) \mapsto x_1 \cdot x_2$ to an edge and get a strictly convex function, we call that edge a

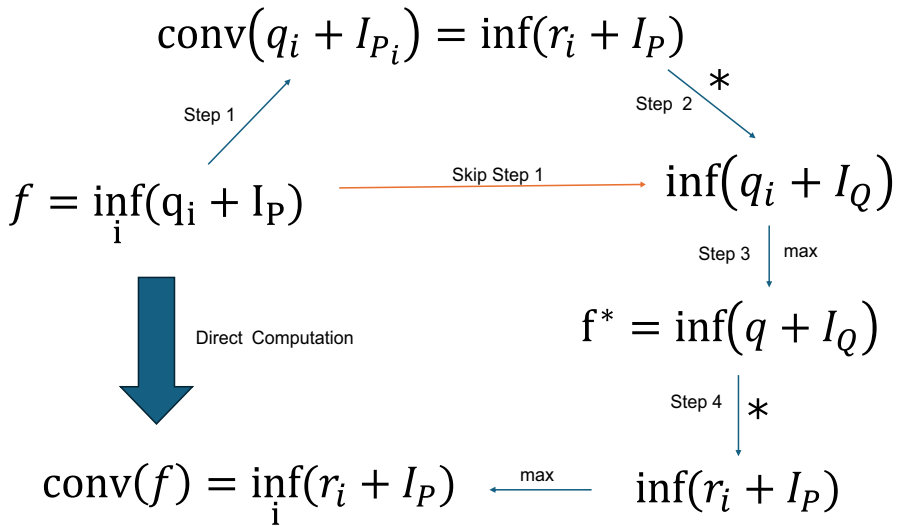


Fig. 1 Description of the process. The computation follows the thin arrows steps 1, 2, 3, 4 and max. Future work will look at direct computation (thick arrow)

convex edge. Each piece (P_i, f_i) of the domain now consists of triangle P_i with either zero, one, two, or three convex edges. To avoid cases with three convex edges, we split such triangles into two subtriangles by adding a horizontal edge through a vertex such that each triangle has two strictly convex edges. Hence, we will only consider triangles with zero, one or two convex edges.

We then use direct formulae, which we developed using the method given in [6], to obtain the convex envelopes of bilinear forms over triangles. The derivation of those formulae is explained in Appendix A.

Kumar [45] showed how to compute the conjugate of a rational function defined on a polytope. We further fine tuned his method to obtain formulae given in Appendix B to compute the conjugate which is Step 2 of the above process. We then computed the maximum of the functions obtained in Step 3 thereby obtaining the conjugate of a bivariate PLQ function.

Beyond simplifying previous work [6, 45], our focus here is to explain how to perform the last step, required to obtain the convex envelope, by computing the conjugate of the conjugate obtained in Step 3 thereby obtaining the biconjugate of a bivariate PLQ function. Figure 1 shows the overall process of computing the biconjugate. As shown in the figure, we can now skip Step 1 as we have direct formulae to get the conjugate of the bilinear form on a triangle.

Our implementation in MATLAB uses symbolic computation and rational numbers to avoid floating-point errors. Using floating-point arithmetic can give rise to partitions of the domain with degenerate subsets, and floating-point coefficients of functions make it difficult to detect equal functions on adjacent domains. Indeed, one of the motivations of [32] is to check that intermediate computations of conjugates are still convex (as they should be without floating point errors). After observing these

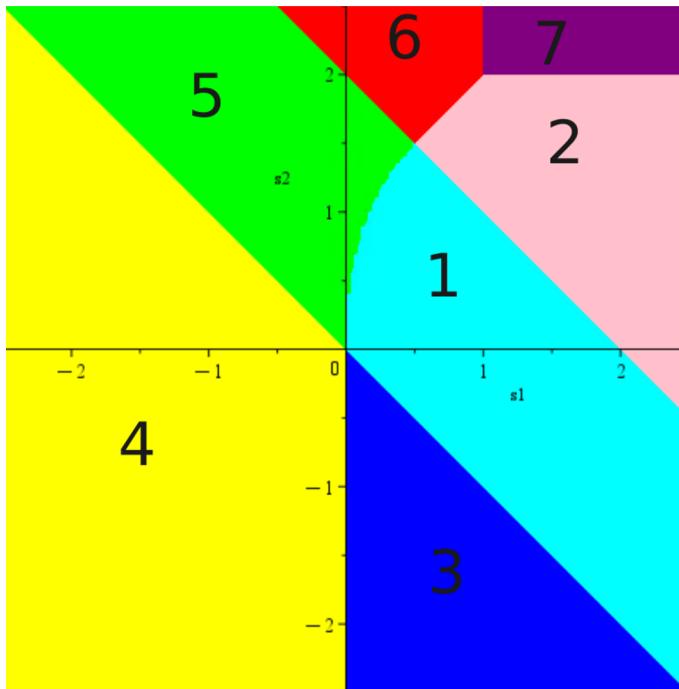


Fig. 2 Domain of the Conjugate of a PLQ function showing a parabolic subdivision. The expression of the functions defined on each piece are provided in Table 1

difficulties, we decided to work with symbolic representation of our functions with rational coefficients.

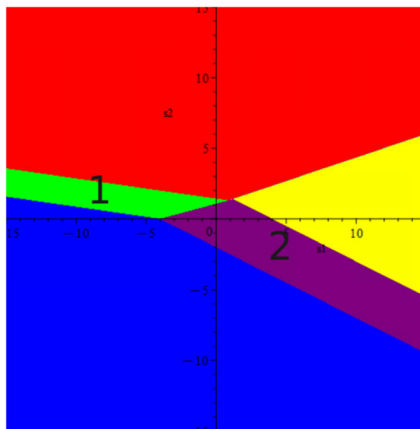
3 Structure of the conjugate of a PLQ function

In order to compute the conjugate of a PLQ function f , we first triangulate each polyhedral set making up its domain. Then compute the convex envelope of the restriction of the function to each triangle. (The computation of the convex envelope of the bilinear function over each triangle is given in Appendix A.) Next we compute the conjugates of each of these convex envelopes as per Appendix B, and then the maximum of all those conjugates to obtain the conjugate f^* . By exhaustively considering all cases we observe a particular form for the conjugate of a PLQ function. Figure 2 shows an example of the domain of a conjugate. We have labeled each region from 1 to 7 and the corresponding functions defined in each region are listed in Table 1. The conjugate of a PLQ function is a piecewise function with four types of pieces:

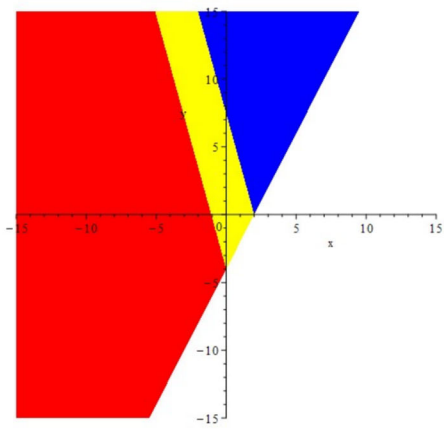
1. Type 1 : Linear functions on a polyhedral subdivision, *e.g.*, pieces 2,3,4,6 and 7 from Figure 2.

Table 1 Function expressions for Fig. 2

Piece number	Expression
1	$2s_1$
2	$2s_1$
3	$2s_1$
4	0
5	$(s_1 + s_2)^2/4$
6	$s_1 + s_2 - 1$
7	$2s_1 + s_2 - 2$



(a) Type 4



(b) Subdifferential of Piece 2

Fig. 3 Domain of the conjugate of a piecewise function along with the subdifferential of one piece.(Functions are defined in Table 2**Table 2** Function expressions for Fig. 3

Piece number	Expression
1	$\frac{1}{8}s_1^2 + \frac{1}{2}s_1s_2 + s_1 + \frac{1}{2}s_2^2 - 2s_2 + 2$
2	$\frac{1}{28}s_1^2 + \frac{1}{2}s_1s_2 + \frac{2}{7}s_1 + \frac{7}{4}s_2^2 - 2s_2 + \frac{4}{7}$

2. Type 2 : Linear function defined on a region with a parabolic edge and two parallel linear edges. This region is the convex side of the parabolic edge (*e.g.*, Piece 1 of Figure 2).
3. Type 3 : Parabolic function defined on the nonconvex side of the parabolic edge between the same two linear parallel edges(*e.g.*, Piece 5 of Figure 2). Note that this region is nonconvex because it obtained as the union of convex sets (rays) by merging convex sets to simplify the function representation.
4. Type 4 : Parabolic function on an unbounded polyhedral subdivision which has three edges, two of which are parallel (*e.g.*, Figure 3).

We exploit this structure to compute the biconjugate of the PLQ function.

We now discuss the computation of the conjugate of a parabolic function defined over a parabolic subdivision. We start with computing the subdifferentials corresponding to each entity of the subdivision and then compute the expressions over them to finally obtain the conjugate. An entity of the parabolic subdivision is either a vertex, an edge or a face of the parabolic subdivision of $\text{dom } f^*$.

4 Domain of the biconjugate

Given a nonconvex PLQ function, we compute the conjugate and obtain a piecewise parabolic function. We now compute the conjugate of each piece of the conjugate by first computing its domain which turns out to be a polyhedral subdivision. (Recall that $\overline{\text{dom } f^*} = \overline{\text{ran } \partial f}$ the closure of the range of ∂f [42, Remark 18.16].) We decompose the domain of f^* (parabolic subdivision) into its interior, its vertices and its edges.

The subdifferential of any point in the interior or any of the vertices is identical for all four types of pieces mentioned in Sect. 3.

Let

$$q_p(x) = ax_1^2 + bx_1x_2 + cx_2^2 + dx_1 + ex_2 + f, \text{ with } a > 0, c > 0, b^2 - 4ac = 0, \quad (1)$$

be a parabolic function. Note that q_p is a convex function (the letter q is used for quadratic, and the subscript p for parabolic).

Proposition 1 (Interior) *Consider q_p defined in (1). Then there exists α_i such that*

$$\bigcup_{x \in \text{dom } q_p} \partial q_p(x) \subseteq \{s : L(s) = 0\},$$

where $L(s) = \alpha_1 s_1 + \alpha_2 s_2 + \alpha_0$.

Proof Take $s \in \bigcup_{x \in \text{dom } q_p} \partial q_p(x)$. There exists x such that $s_1 = 2ax_1 + bx_2 + d$ and $s_2 = bx_1 + 2cx_2 + e$. Define $\alpha_1 = b$, $\alpha_2 = -2a$, $\alpha_0 = -db + 2ae$, and substitute values of s_1 and s_2 to obtain

$$\alpha_1 s_1 + \alpha_2 s_2 + \alpha_0 = (b^2 - 4ac)x_2.$$

Using $b^2 - 4ac = 0$ finishes the proof. $\square$

Remark 1 Proposition 1 is more general than [36, Proposition 1] as a polyhedral subdivision is a special case of a parabolic subdivision.

Corollary 2 (Interior) *For a bivariate parabolic function q_p as defined in (1), and a set P belonging to a parabolic subdivision P_a , define $f(x) = q_p(x) + I_P(x)$, then for all $x \in \text{int}(P)$, the set $\bigcup_{x \in \text{int}(P)} \partial f(x)$ is contained in a line.*

Proof For any point $x \in \text{int } P$, we have $\partial f = \partial q$ so the result follows by Proposition 1. $\square$

Example 1 For the parabolic function, $q(s) = (s_1 + s_2)^2/4$ defined in the interior of a parabolic division as shown in Figure 2, the subdifferential is contained in a line segment $\{x : x_2 = x_1\}$ between $(0, 0)$ and $(1, 1)$ as shown in Figure 6[e].

Proposition 3 (Vertices) *For a parabolic function q_p as defined in (1), a parabolic unbounded region P , and a vertex v of P . Let $f(x) = q_p(x) + I_P(x)$. Then $\partial f(v)$ is an unbounded polyhedral set.*

Proof Let v be a vertex of P and E^- and E^+ be the tangents to the edges at the vertex v . Then for all $y \in P$, $y_l \in E^-$ and $y_r \in E^+$, the normal cone at v is

$$\begin{aligned} N_P(v) &= \{s \in \mathbb{R}^2 : s^T(y - v) \leq 0, \forall y \in P\}, \\ &= \{s_l \in \mathbb{R}^2 : s_l^T(y_l - v) \leq 0\} \cap \{s_r \in \mathbb{R}^2 : s_r^T(y_r - v) \leq 0\}, \\ &= \{s \in \mathbb{R}^2 : s^T(y_l - v) \leq 0, s^T(y_r - v) \leq 0\}, \end{aligned} \quad (2)$$

which is an unbounded polyhedral set. Therefore the subdifferential given by

$$\begin{aligned} \partial f(v) &= \partial q_p(v) + N_P(v), \\ &= \{s + \nabla q_p(v) \in \mathbb{R}^2 : s^T(y_l - v) \leq 0, s^T(y_r - v) \leq 0\}, \end{aligned} \quad (3)$$

is also an unbounded polyhedral set. $\square$

Remark 2 The polyhedrality of the normal cone can also be deduced from [15, Theorem 6.14] and the subsequent note on p. 211 that states that the normal cone is generated by gradients of the active constraints. Since there are a finite number of inequalities and all functions are differentiable, there is a finite number of such gradients, which makes the normal cone polyhedral.

Remark 3 Since linear functions are a subset of parabolic functions, our result generalizes [36, Lemma 1].

Example 2 Consider piece P_1 shown in Figure 2

$$P_1 = \{(s_1, s_2) : -s_1 - s_2 \leq 0, s_1 + s_2 - 2 \leq 0, -8s_1 + 2s_1s_2 + s_1^2 + s_2^2 \leq 0\}.$$

The function $f(s_1, s_2) = 2s_1$ is defined on P_1 . At the vertex $V = (0, 0)$, $\nabla q(V) = (2, 0)$ and the normal cone is $N_{P_1}(V) = \{x : x_2 \leq 0, x_1 - x_2 \leq 0\}$ shown in Figure 4. The subdifferential is given by $\partial f_{P_1}(V) = \{(2, 0) + (x_1, x_2), x_2 \leq 0, x_1 - x_2 \leq 0\}$.

Subdifferentials computed at the vertices of a polyhedral regions are given in [36] and can also be considered as a particular case of Proposition 3 as a polyhedral division is a special type of parabolic division.

For computing the subdifferentials over the edges, we consider all the cases listed in Table 3.

Lemma 4 (Linear edge and a linear function [45, Corollary 4.5]) *Let l be a linear function defined on a linear edge $E = \{x : x_2 = mx_1 + c, x_{l_1} \leq x_1 \leq x_{u_1}\}$ (x_{l_1} or x_{u_1} maybe ∞). Then $\partial l(E)$ is a singleton.*

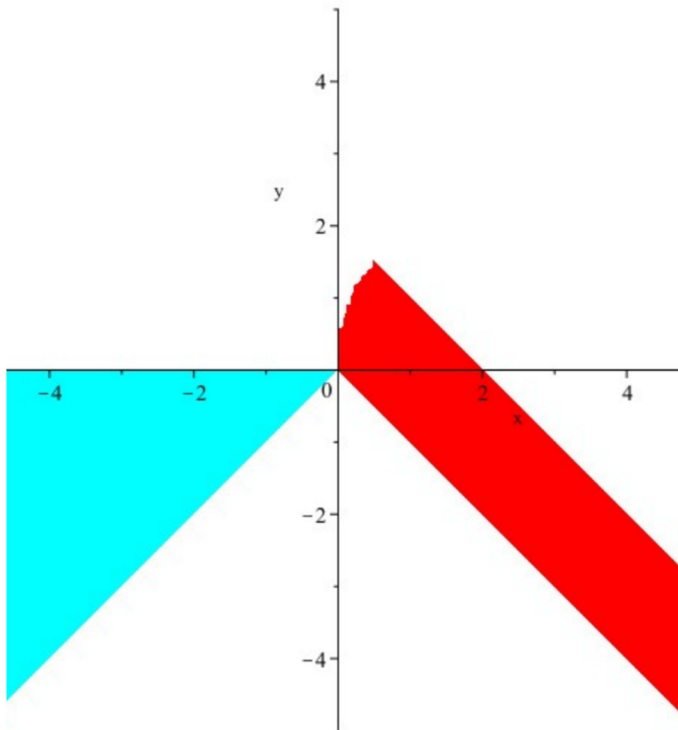


Fig. 4 Piece 1 - Vertex 1. Normal Cone at Vertex $v = (0, 0)$

Table 3 Subdifferential based on edges

Type of piece	Function	Edge	Lemma	Subdifferential
1, 2	Linear	Linear	4	Singleton
2	Linear	Parabolic	8	Cone
3, 4	Parabolic	Linear(Parallel)	6	Ray
3	Parabolic	Parabolic	7	Polyhedral region
4	Parabolic	Linear	7	Polyhedral region

Proposition 5 (Linear edge and a parabolic function.) *Take q_p as in (1), $P \in P_a$ a parabolic region, and $L = \{x : x_2 - mx_1 - c = 0, x_{l_1} \leq x_1 \leq x_{u_1}\}$ a linear edge. Then for any $x \in L$ and $f(x) = q_p(x) + I_P(x)$, $\partial f(x) \subseteq \{\hat{s} + \hat{s} : b\hat{s}_1 - 2a\hat{s}_2 - bd + 2ae = 0, \hat{s}_1 = -m\hat{s}_2, \hat{s}_2 \geq 0\}$ is a polyhedral region or a ray.*

Proof For $x \in \text{ri}(L)$,

$$\begin{aligned}
 N_P(x) &= \{\hat{s} : \hat{s} = \lambda(-m, 1), \lambda \geq 0\}, \\
 &= \{\hat{s} : \hat{s}_1 = -m\hat{s}_2, \hat{s}_2 \geq 0\}.
 \end{aligned} \tag{4}$$

From Proposition 1 we have

$$\bigcup_{x \in \text{dom } q_p} \partial q_p(x) \subseteq \{\bar{s} : b\bar{s}_1 - 2a\bar{s}_2 - bd + 2ae = 0\}.$$

Now for all $x \in \text{ri}(L)$, $\partial f(x) = \partial q_p(x) + N_P(x)$, so

$$\partial f(x) \subseteq \{\bar{s} + \hat{s} : b\bar{s}_1 - 2a\bar{s}_2 - bd + 2ae = 0, \hat{s}_1 = -m\hat{s}_2, \hat{s}_2 \geq 0\},$$

which concludes the proof. $\square$

Corollary 6 For a parabolic function q_p restricted to a linear edge L of a piece of Type 3, $\partial f(x)$ is contained in a ray.

Proof We get pieces of Type 3 by taking the conjugate of a rational function restricted to a convex edge. The linear edges are parallel to the normal cone obtained at the convex edge. The slope of a parallel edge of f^* satisfies $m_1 = -1/m$, where m is the slope of the convex edge of f and m_1 is the slope of the edge L . The parabolic function computed for this type is given by $q_p = ax^2 + bxy + cy^2 + dx + ey + f$, where $a = 1/(4m)$, $b = 1/2$, $c = m/4$, $d = -q/(2m)$, $e = q/2$, $f = q^2/(4m)$; see Appendix B. Substituting the value for m_1 , and the values for the coefficients in $\partial f(x)$ obtained in Proposition 5, we get

$$\partial f(x) \subseteq \{\bar{s} + \hat{s} : m\bar{s}_1 - \bar{s}_2 + q = 0, \hat{s}_2 = m\hat{s}_1, \hat{s}_2 \geq 0\}.$$

This gives us a ray. $\square$

Example 3 Consider piece P_5 shown in Figure 2

$$P_5 = \{(s_1, s_2) : -s_1 - s_2 \leq 0, s_1 + s_2 - 2 \leq 0, 8s_1 - 2s_1s_2 - s_1^2 - s_2^2 \leq 0\}.$$

The function $f(s_1, s_2) = (1/4)(s_1 + s_2)^2$ is defined on this region. At the edge, $s_1 + s_2 = 0$ which is the ray from vertex, $V_1(0, 0)$, $\nabla f(s_1, s_2) = (1/2)(s_1, s_2)$ and the subdifferential is contained in the ray $\{s : s_2 - s_1 = 0, s_2 \leq 0\}$.

Corollary 7 For a parabolic function q_p restricted to a linear edge (not parallel) of a piece of Type 4, $\partial f(x)$ is an unbounded polyhedral region.

Example 4 Referring to the purple piece in Figure 3, corresponding to the edge between the purple and green piece we get the yellow region as shown in the subdifferential in Figure 3.

Remark 4 The pieces of “Type 3” are nonconvex, joining the two vertices with a line gives us its convex envelope. We compute the normal cone of the convex envelope in order to compute the domain of the conjugate. We use Corollary 7 to compute the normal cone.

Proposition 8 (Parabolic edge and linear function) *For a linear function l , restricted to the parabolic edge $C(x) = \{x : ax_1^2 + bx_1x_2 + cx_2^2 + dx_1 + ex_2 + f, b^2 - 4ac = 0, x_l \leq x_1 \leq x_r\}$, then $\partial f(E)$ is a cone with vertex at $\nabla l(x)$.*

Proof Let $g(x)$ be the restriction of a linear function to the parabolic curve joining x_l and x_r be given by

$$C(x) = ax_1^2 + bx_1x_2 + cx_2^2 + dx_1 + ex_2 + f, b^2 - 4ac = 0. \quad (5)$$

For $s \in \bigcup_{x \in \text{dom } g} \partial g(x)$, $s = \lambda \nabla C(x)$, $\lambda \geq 0$. This gives us

$$s_1 = \lambda(2ax_1 + bx_2 + d), s_2 = \lambda(bx_1 + 2cx_2 + e). \quad (6)$$

Solving equations (5) and (6) for x_1, x_2 and λ and substituting, we get the normal cone between rays, $s_2 = k_l s_1$ and $s_2 = k_r s_1$, where $k_l = (e + bx_l + 2cy_l)/(d + 2ax_l + by_l)$ and $k_r = (e + bx_r + 2cy_r)/(d + 2ax_r + by_r)$. For all $x \in \text{ri}(C)$, $\partial f(x) = \partial g(x) + N_P(x)$, thus

$$\partial f(x) = \{\nabla g(x) + (s_1, s_2), k_l s_1 \leq s_2 \leq k_r s_1\}.$$

□

Example 5 Consider piece P_1 shown in Figure 2

$$P_1 = \{(x, y) : -x - 2y - 4 \leq 0, x + 2y - 4 \leq 0, 48x - 56y + 4xy + x^2 + 4y^2 - 184 \leq 0\}.$$

The function $f(x, y) = 2x$ is defined on P_1 . At the edge between vertex $V_1(0, 0)$ and vertex $V_2(0.5, 1.5)$, $\nabla l(v) = (2, 0)$ and the subdifferential is described by $-s_2 \leq 0$ and $s_1 + s_2 - 2 \leq 0$ as shown in Figure 5.

Locatelli [6] showed that $\text{conv } f$ is a piecewise function over a polyhedral subdivision that can be linear, quadratic, or rational. By gathering Propositions 1, 3, 5 and 8, Corollaries 2, 6 and 7 and Lemma 4 we get the same theorem with a constructive proof.

Theorem 9 (Polyhedral domain) *For a piecewise quadratic-linear function, the domain of the biconjugate has a polyhedral subdivision.*

5 Biconjugate expression

Now that we know $\text{dom } f^{**}$ as a polyhedral subdivision, we turn to the computation of its expression on each entity.

The conjugate expressions we obtain are either linear functions

$$ax + by + c, \quad (7)$$

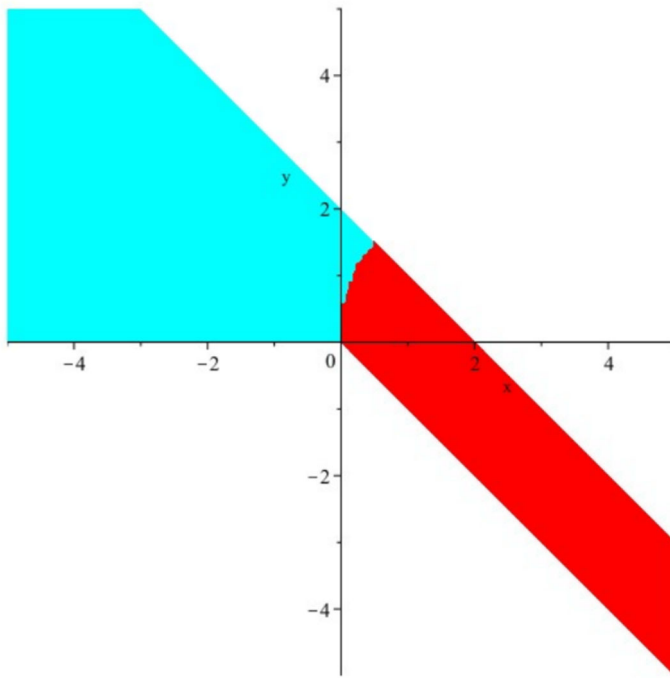


Fig. 5 Subdifferential of $f = 2x$ on the parabolic edge of P_1

parabolic functions

$$ax^2 + bxy + cy^2 + dx + ey + f, a, c > 0, b^2 - 4ac = 0 \quad (8)$$

or a rational functions of the form $\xi_1(x)^2/\xi_2(x) + \xi_0(x)$ where ξ_1, ξ_2 and ξ_0 are linear functions. These rational functions only occur on an edge of the triangle where the function restricted to that edge is convex. (That edge cannot be vertical because then the function $f(x) = x_1 \cdot x_2 = 0$.) Let the convex edge be $y = mx + q$ and the vertex opposite it be (x_1, y_1) , then the rational expression can be written as

$$r(x, y) = \frac{ax^2 + bxy + cy^2 + dx + ey + f}{gx + hy + k}, \quad (9)$$

where $a = -my_1, b = q, c = x_1, d = -qy_1 + mx_1y_1, e = -qx_1 - x_1y_1, f = qx_1y_1, g = -m, h = 1$, and $k = -y_1 + mx_1$ (16).

Note that (8) and (7) are special cases of (9).

Theorem 10 For a PLQ function, the biconjugate has a polyhedral subdivision with a rational function of the form given in (9).

Proof We are computing the expressions on nonempty regions as listed in Table 4 and then computing the maximum over all the regions.

Table 4 Conjugate expressions and domain types for each piece type with associated cases as indicated in the proof of Theorem 10

Type of piece	Function	Vertex	Domain	Expression	Case
1, 2, 3, 4	Linear/Parabolic	Linear	Polyhedral region	Linear	1
Type of piece	Function	Edge	Domain	Expression	Case
2	Linear	Parabolic	Cone	Rational function	3
3	Parabolic	Parabolic	Polyhedral region	Rational function	2
4	Parabolic	Linear	Polyhedral region	Parabolic function	4

Case 1 (Vertices) Let $V \subset P$ be the set of all vertices. Then for each $v \in V$

$$\begin{aligned} f^*(s) &= \max_{x \in \{v\}} (s^T x - f(x)), \\ &= \max_{x \in \{v\}} (s^T x - (q(x) + I_P(x))), \\ &= s^T v - q(v), \\ &= s_1 v_x + s_2 v_y - q(v), \end{aligned}$$

is a linear function. So for any vertex v , the conjugate is a linear function defined on an unbounded polyhedral set.

Case 2 (Edge:parabolic; Function:parabolic)

When the edge is parabolic it is of a particular form; $edge = ax_1^2 + bx_1x_2 + cx_2^2 + dx_1 + ex_2 + f$, where $a = -1$, $b = -2m$, $d = 2q + 4mx_1$, $c = -m^2$, $e = -(2mq - 4my_1)$ and $f = -(q^2 + 4mx_1y_1)$ and the conjugate function is $q_p = a_1x_1^2 + b_1x_1x_2 + c_1x_2^2 + d_1x_1 + e_1x_2 + f_1$ where $a_1 = \frac{1}{4m}$, $b_1 = \frac{1}{2}$, $c_1 = \frac{m}{4}$, $d_1 = -\frac{q}{2m}$, $e_1 = \frac{q}{2}$, $f_1 = \frac{q^2}{4m}$ as given in Appendix B.

As the conjugate expression is defined as $f^*(s) = \sup_{x \in E} (s^T x - f(x))$ and the function q_p and $edge$ are differentiable,

we can compute the supremum to get

$$f^*(s) = \frac{a_2s_1^2 + b_2s_1s_2 + c_2s_2^2 + d_2s_1 + e_2s_2 + f_2}{g_2s_1 + h_2s_2 + k_2}, \quad (10)$$

where $a_2 = -my_1$, $b_2 = q$, $c_2 = x_1$, $d_2 = -qy_1 + mx_1y_1$, $e_2 = -qx_1 - x_1y_1$, $f_2 = qx_1y_1$, $g_2 = -m$, $h_2 = 1$, and $k_2 = -y_1 + mx_1$. (Expressions are computed symbolically using the code give in Appendix C.1.)

Case 3 (Edge:parabolic; Function:linear) In this case, we get the conjugate expression as a rational function given in (10). We compute the conjugate expression using the code given in the Appendix C.1 by passing the parabolic edge and a linear function as input.

Case 4 (Edge:linear; Function:parabolic) When the edge is linear, $y = mx + q$ and the function is quadratic, $q = ax^2 + bxy + cy^2 + dx + ey + f$, we get the conjugate expression as a parabolic function. Again this was computed using the code given in the Appendix C.1. $\square$

All the cases are verified symbolically and the code used to compute the conjugate expressions is given in Appendix C.1.

Computing the maximum over all the pieces, we obtain a rational function of the forms (9) on a polyhedral subdivision.

6 Complexity of the algorithm

Our algorithm runs in linear worst-case time.

Theorem 11 *Our algorithm to compute the biconjugate of a PLQ function f defined on a polyhedral subdivision with n vertexes runs in $O(n)$ worst-case time.*

Proof Denote f a PLQ function defined on a polyhedral subdivision with n vertexes. We obtain the convex envelope of a PLQ function by following five steps.

Step 1: We start by triangulating each piece, which takes $O(n)$ since triangulating convex polyhedral sets takes linear time [35, p. 49] (pick a vertex and join it to all other vertexes). After triangulating we will have m edges where $m < 3n$. The total number of common edges between any two pieces would be $n_c = m - n$ and the number of triangle is $t \leq n - 1$.

Step 2: Each restriction of f on a triangle is quadratic. Using a rotation if necessary, it becomes either a convex quadratic, a concave quadratic or the function $(x, y) \mapsto xy$. The first case is convex so it equals its convex envelope. The convex envelope of a concave quadratic defined on a triangle with vertexes v_i is a piecewise linear function going through the points $(v_i, f(v_i))$, which is easily computed in constant time. We provide an explicit formula to compute the convex envelope of the bilinear function over a triangle and obtain it in $O(1)$ time.

Step 3: We now compute the conjugate of each convex envelope of the restriction of f to a triangle. Computing normal cones provide the domain of the conjugate, and we have a direct formula for the expressions corresponding to each entity. The maximum number of entities for each triangle is six. Thus the complexity for this step is $O(t)$, where t is the number of triangles.

Step 4: Next we compute the maximum of the conjugates over the entire PLQ function to obtain the conjugate of the PLQ function. As given in [37], the complexity of computing the maximum is $O(n_c)$, where n_c is the number of common edges in the original triangulation. We have now obtained the conjugate f^* of f .

Step 5: Finally we compute the biconjugate $f^{**} = (f^*)^*$ by computing the conjugate of f^* . We note that the restriction of f^* to a triangle is already convex (no need to compute the convex envelope of each piece), and f^* is not defined on a polyhedral subdivision, but on a parabolic subdivision. Adapting our previous computation, the computation is a constant multiple of the number of common edges. Since the number of common edges is less than $2n$, we get a total complexity of $O(n)$.

Adding all computation time together, we obtain $O(n)$ worst-case time complexity. $\square$

The next section provides several illustrative examples.

7 Examples

In this section a few examples of the conjugates and biconjugates of PLQ functions are given. Example 6 is a simple example where the input PLQ function domain is partitioned into seven pieces; we find that its biconjugate has a domain partitioned into two pieces. Example 7 contains a PLQ whose domain is partitioned in more pieces.

Example 6 The conjugate shown in Figure 2 has seven pieces, three vertices, one quadratic edge between the first two vertices, one linear edge (line) and some rays.

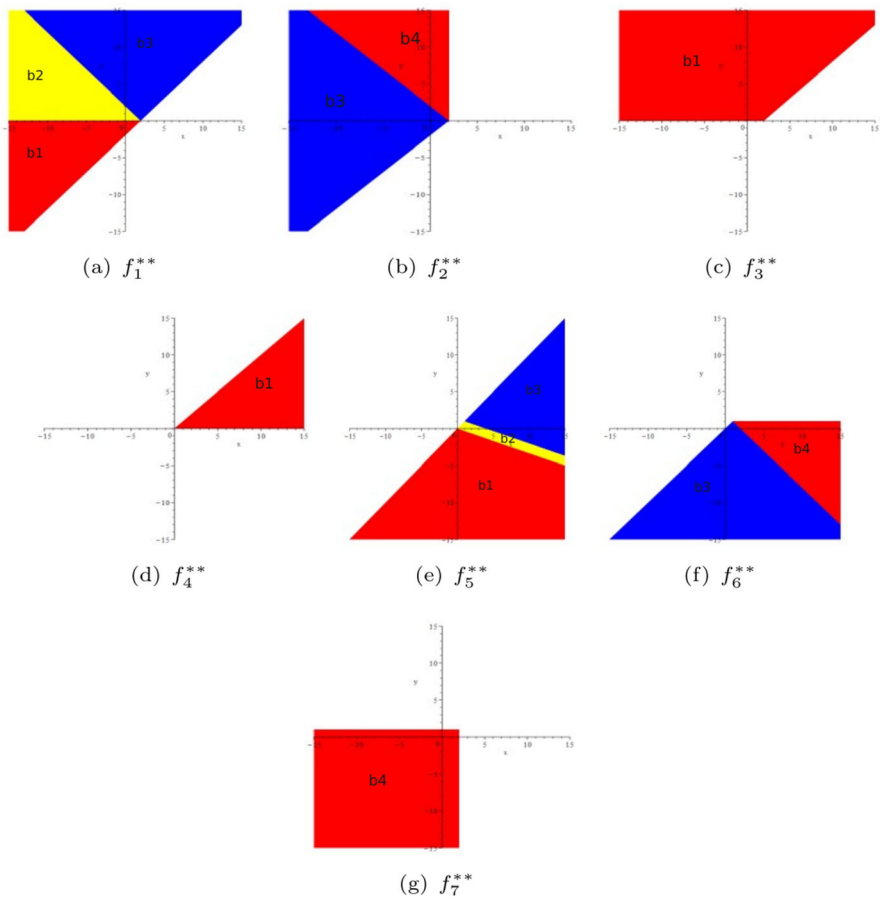


Fig. 6 Domain of the biconjugate f_i^{**} of the conjugate function shown in Figure 2. The associated functions are listed in Table 5

In order to compute its biconjugate, we first compute the conjugate of each piece and then take the maximum. The biconjugates of each piece (1-7) of the conjugate are shown in Figure 6.

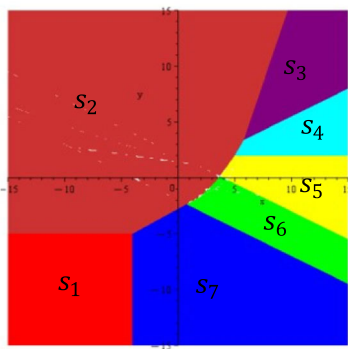
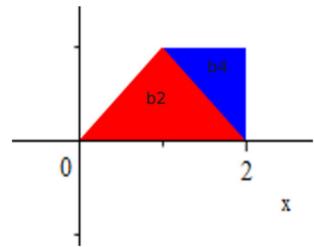
The first piece of the conjugate is the function $(s_1, s_2) \mapsto 2s_1$ defined over the domain given by the inequalities $-s_1 - s_2 \leq 0$, $2s_1s_2 - 8s_1 + s_1^2 + s_2^2 \leq 0$ and $s_1 + s_2 - 2 \leq 0$, and vertices $(0, 0)$ and $(1/2, 3/2)$ (it is marked “1” in Figure 2). We compute its biconjugate f_1^{**} (its domain is shown in Figure 6(a)). Similarly the biconjugates of pieces 2-7 are shown in figures 6(b)–(g). We then take the maximum of these functions over the intersecting domains to get the biconjugate of the original PLQ function which is shown in Figure 7.

Example 7 In our earlier work [37] we had an example where we have computed the conjugate of a PLQ function. The conjugate is shown in Figure 8[a]. We have now computed the biconjugate of this function and it is shown in Figure 8[b]. The Fig-

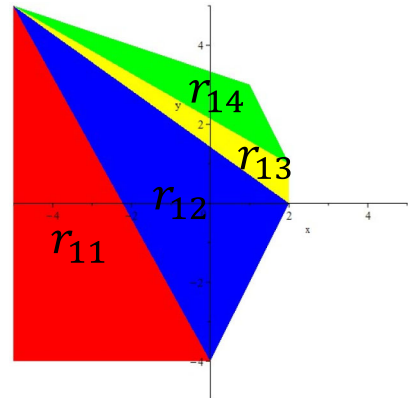
Table 5 Function definitions for Fig. 6

Piece number	Expression
b1	0
b2	$2y^2/(y-x+2)$
b3	$x/2 + 3y/2 - 1$
b4	$x + 2y - 2$

Fig. 7 Domain of the biconjugate of the PLQ function plotted in Figure 2; its expression on each partition is listed Table 5. It is obtained as the maximum $f^{**} = \max_i (f_i^{**})$ of each function shown in Figure 6



(a) Conjugate



(b) Biconjugate

Fig. 8 Conjugate and Biconjugate of Example 7. (The functions are enumerated in the Table 6)

ures show the domains while the conjugate functions $s_1, \dots, s_7$ and the biconjugates (rational functions) are given in the Table 6. In this example the polyhedral division obtained after computing the biconjugate is different from the polyhedral division of the original PLQ function.

Remark 5 In Example 6 the polyhedral division of the original PLQ function and the biconjugate is identical. Thus the functions computed in Step1 by direct formulae are the same ones obtained after computing the biconjugate. Whereas in Example 7 we get a different polyhedral division. The biconjugate functions we obtain here match the once we get in Step 1 if we had started with the same division. This indicates that we might be able to compute the biconjugate functions by only computing the polyhedral division of the biconjugate.

Table 6 Function definitions for Fig. 8

Function name	Expression
r_{11}	$-4x - 5y - 20$
r_{12}	$(30x - 5y + 4xy + 10x^2 + 5y^2 - 100)/(2x - y + 15)$
r_{13}	$5x + 2y - 10$
r_{14}	$29x/5 + 17y/5 - 13$
s_1	$-5s_1 + 4s_2 - 20$
s_2	$-5s_1 + 5s_2 + 25$
s_3	$s_1 + 3s_2 - 3$
s_4	$2s_1 + s_2 - 2$
s_5	$2s_1$
s_6	$s_1^2/8 + s_1s_2/2 + s_1 + s_2^2/8 - 2s_2 + 2$
s_7	$-4s_2$

8 Conclusion

Previous work showed a linear-time algorithm to compute the conjugate of a bivariate PLQ function. Here we show a linear time algorithm to compute the biconjugate thereby the conjugate and biconjugate of a bivariate PLQ function can be obtained in linear time.

While it was known that the biconjugate admits a polyhedral subdivision, its expression was unknown. We show that it is a rational function.

As future work, once we have the subdivision of the domain of the biconjugate, we can directly compute the expressions using the same formulae obtained in A.

After observing experimental results, it seems possible to predict the subdivision of the domain of the biconjugate from the domain of the original PLQ function. This seems possible when the original domain is convex and connected. Experimentally, there seem to be patterns emerging in the domain of the biconjugate, hence the next step would be to compute these divisions directly. If we can get direct formulae, then it might be possible to recognize patterns and extend the work to three or more variables.

Another possible direction is to explore non-convex domains for PLQ functions.

From a coding perspective, the code could be parallelized resulting in faster computation.

From an application perspective, the convex envelope is used to determine lower bounds while looking for a global minimum, for example in a branch-and-bound algorithm. It would be interesting to look into whether having explicit expressions that are quickly computed help such computation.

Appendices

A Convex envelope of a bilinear function over a triangle

When a function is restricted to an edge of the triangle and the resulting function is strictly convex, we call the edge a convex edge. In case of bilinear functions, when the slope is positive, the edge is convex. The triangular pieces can be classified by the number of convex edges they have. Thus we have the following triangles:

1. No Convex edges.
2. One Convex edge.
3. Two Convex edges.
4. Three Convex edges.

The paper [6] gives us a method to compute the convex envelope of a bilinear function on a polytope. We have solved these optimization problems symbolically to obtain direct formulae to obtain the convex envelopes. This gives us the envelopes in one step without having to solve an exponential number of optimization problems. We provide the formulae in this section using the same notations as [6].

A.1 Notation

Definition 1 The convex envelope is defined as the pointwise supremum of the under-estimating affine functions, f over P . Given $(x, y) \in P$, the value of the convex envelope of f over P is the optimal value of the optimization problem in the three variables a, b and c with an infinite number of constraints.

$$\text{Conv}_{f,P}(x, y) = \max_{a,b,c} \{ c \mid f(x', y') - [a(x' - x) + b(y' - y) + c] \geq 0, \forall (x', y') \in P \} \quad (11)$$

The optimal solution (a^*, b^*, c^*) of this problem gives a supporting hyperplane for the convex envelope. In [6] it is shown that the problem can be rewritten as

$$\text{Conv}_{f,P}(x, y) = \max \{ c \mid f(x_{v_i}, y_{v_i}) - [a(x_{v_i} - x) + b(y_{v_i} - y) + c] \geq 0, \forall i \in V_P, \\ \min_{(x', y') \in [v_{j_1}, v_{j_2}]} f(x', y') - [a(x' - x) + b(y' - y) + c] \geq 0, \forall j \in E_P \},$$

where V_P is the set of vertices which do not belong to any edge in E_P (each vertex is denoted by (x_{v_i}, y_{v_i}) , and E_P is the set of indices of the edges along which f is strictly convex. (For each $j \in E_P$, we have an edge E_j defined by its end points (Vertices v_{j_1} and v_{j_2}); we denote each edge by a line $y' = m_j x' + q_j$ and if we restrict f to this edge we get $f(x') = f(x', m_j x' + q_j)$.)

Now the constraint related to edge $j \in E_P$ with $E_j = [v_{j_1}, v_{j_2}]$ can be written as

$$\min_{x' \in [x_{v_{j_1}}, x_{v_{j_2}}]} \{ f(x') - [a(x' - x) + b((m_j x' + q_j) - y)] \} \geq c, \quad (12)$$

where $x_{v_{j_1}}, x_{v_{j_2}}, x_{v_{j_1}} \leq x_{v_{j_2}}$ are the x -coordinates respectively of the vertices v_{j_1} and v_{j_2} respectively.

Let $s_j(a, b)$ be the unconstrained minimum point of $x' \mapsto f_j(x') - (a + bm_j)x'$.

If $s_j(a, b) \neq \pm\infty$ then $f'_j(s_j(a, b)) = a + bm_j$.

Let us now introduce three regions :

$$D_j^- = \{(a, b) : s_j(a, b) \leq x_{v_{j_1}}\}$$

$$D_j^+ = \{(a, b) : s_j(a, b) \geq x_{v_{j_2}}\}$$

$$D_j = \{(a, b) : x_{v_{j_1}} \leq s_j(a, b) \leq x_{v_{j_2}}\}$$

As f is strictly convex along the edge, the first derivative is increasing in the interval $[x_{v_{j_1}}, x_{v_{j_2}}]$, we have

$$D_j^- = \{(a, b) : a + bm_j \leq f'_j(x_{v_{j_1}})\}$$

$$D_j^+ = \{(a, b) : a + bm_j \geq f'_j(x_{v_{j_2}})\}$$

$$D_j = \{(a, b) : f'_j(x_{v_{j_1}}) \leq a + bm_j \leq f'_j(x_{v_{j_2}})\}$$

Therefore for each $j \in E_P$ the optimal solution is

$$x_j(a, b) = \begin{cases} x_{v_{j_1}} & : \text{if } (a, b) \in D_j^- \\ x_{v_{j_2}} & : \text{if } (a, b) \in D_j^+ \\ s_j(a, b) & : \text{if } (a, b) \in D_j \end{cases}$$

and its corresponding y coordinate is

$$y_j(a, b) = m_j x_j(a, b) + q_j.$$

For each i in V_P we have $x_i(a, b) = x_{v_i}$, $y_i(a, b) = y_{v_i}$

So if we define a function

$$\eta_k(a, b) = f(x_k(a, b), y_k(a, b)) - ax_k(a, b) - by_k(a, b), \forall k \in V_P \cup E_P; \quad (13)$$

then we can finally write (11) as

$$\text{Conv}_{f,P}(x, y) = \max\{c \mid \eta_k(a, b) + ax + by \geq c, \forall k \in V_P \cup E_P\}. \quad (14)$$

A.2 No convex edges

When there are no convex edges, the set of convex edges is empty while the set of vertices has all three vertices.

A.2.1 Optimization problem

Subproblems for every pair $(i, j), i, j \in V_P, i \neq j$ are

$$\begin{aligned} \max\{\eta_i(a, b) + ax + by : \eta_i(a, b) = \eta_j(a, b), \\ \eta_i(a, b) \leq \eta_k(a, b), k \neq i, k \neq j, \\ (a, b) \in S_r\}. \end{aligned}$$

In this case, we have three optimization problems for vertex pairs (1,2), (1,3) and (2,3) as follows

$$\max\{\eta_i(a, b) + ax + by : \eta_i(a, b) = \eta_j(a, b)\}.$$

The solutions to all of these is the same as symbolic verified in “LinearlinearT.m”.

The convex envelope for a bilinear form defined on a triangle with vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ is given by $ax + by + c$ where

$$\begin{aligned} a &= \frac{(x_1 y_1 y_2 - x_1 y_1 y_3 - x_2 y_1 y_2 + x_2 y_2 y_3 + x_3 y_1 y_3 - x_3 y_2 y_3)}{(x_1 y_2 - x_2 y_1 - x_1 y_3 + x_3 y_1 + x_2 y_3 - x_3 y_2)}, \\ b &= \frac{(x_1 x_2 y_2 - x_1 x_2 y_1 + x_1 x_3 y_1 - x_1 x_3 y_3 - x_2 x_3 y_2 + x_2 x_3 y_3)}{(x_1 y_2 - x_2 y_1 - x_1 y_3 + x_3 y_1 + x_2 y_3 - x_3 y_2)}, \\ c &= \frac{(x_1 x_2 y_1 y_3 - x_1 x_3 y_1 y_2 - x_1 x_2 y_2 y_3 + x_2 x_3 y_1 y_2 + x_1 x_3 y_2 y_3 - x_2 x_3 y_1 y_3)}{(x_1 y_2 - x_2 y_1 - x_1 y_3 + x_3 y_1 + x_2 y_3 - x_3 y_2)}. \end{aligned}$$

Example 8 The Convex Envelope of $f(x, y) = xy$ over $\text{conv}\{(1, -1), (-1, -1), (-1, 1)\}$ is $(x, y) \mapsto -x - y - 1$ on the same domain.

A.3 One convex edge

Here we have one convex edge $y = mx + q$, hence the set E_P has one edge and the set V_P has one vertex, $V(x_1, y_1)$. Using this information and referring to [6] we solve the following optimization problems in order to obtain the convex envelope.

Subproblems for every pair $(i, j), i, j \in E_P \cup V_P, i \neq j$ are given as

$$\begin{aligned} \max\{\eta_i(a, b) + ax + by : \eta_i(a, b) = \eta_j(a, b), \\ \eta_i(a, b) \leq \eta_k(a, b), k \neq i, k \neq j, \\ (a, b) \in S_r\}. \end{aligned}$$

The η functions obtained are:

- For V , $\eta_w = f(x_1, y_1) - ax_1 - by_1 = x_1 y_1 - ax_1 - by_1$.
- For E , $\eta_h = -(a + mb - q)^2 / (4m) - bq$ for edge $y = mx + q$.
- For E^- , $\eta_h^- = f(x_2, y_2) - ax_2 - by_2 = x_2 y_2 - ax_2 - by_2$.
- For E^+ , $\eta_h^+ = f(x_3, y_3) - ax_3 - by_3 = x_3 y_3 - ax_3 - by_3$.

Thus we have three optimization problems corresponding to $V - E^-$, $V - E$ and $V - E^+$.

A.3.1 Optimization problem (V - E)

In this case, $\eta_h = -\frac{(a+mb-q)^2}{4m} - bq$ corresponding to the edge, while $\eta_w = x_1y_1 - ax_1 - by_1$ corresponding to the vertex (x_1, y_1) . Thus the optimization problem can be written as

$$\max\{\eta_w + ax + by : \eta_w = \eta_h, (a, b) \in S_r\}. \quad (15)$$

By solving this problem symbolically we obtain the objective as a rational function, given by

$$r(x, y) = \frac{ax^2 + bxy + cy^2 + dx + ey + f}{gx + hy + k}, \quad (16)$$

where $a = -my_1$, $b = q$, $c = x_1$, $d = -qy_1 + mx_1y_1$, $e = -qx_1 - x_1y_1$, $f = qx_1y_1$, $g = -m$, $h = 1$ and $k = -y_1 + mx_1$; or we get

$$r(x, y) = \frac{\xi_1(x, y)^2}{\xi_2(x, y)} + \xi_0(x, y)$$

where $\xi_1(x, y) = a_1x + b_1y + c_1$ with

$$\begin{aligned} a_1 &= \frac{q - 2y_1}{4(q - y_1 + mx_1)}, \\ b_1 &= \frac{q + 2mx_1}{4m(q - y_1 + mx_1)}, \\ c_1 &= \frac{q(-y_1 - mx_1)}{4m(q - y_1 + mx_1)}; \end{aligned}$$

$\xi_2(x, y) = a_2x + b_2y + c_2$ with

$$\begin{aligned} a_2 &= \frac{-1}{4(q - y_1 + mx_1)}, \\ b_2 &= \frac{1}{4m(q - y_1 + mx_1)}, \\ c_2 &= \frac{(-y_1 + mx_1)}{4m(q - y_1 + mx_1)}; \end{aligned}$$

and $\xi_0(x, y) = a_0x + b_0y + c_0$ with

$$\begin{aligned} a_0 &= \frac{q^2 + 4mx_1y_1}{4(q - y_1 + mx_1)}, \\ b_0 &= \frac{-q^2 - 4mx_1y_1}{4m(q - y_1 + mx_1)}, \\ c_0 &= \frac{(q^2y_1 - mq^2x_1 + 4mqx_1y_1)}{4m(q - y_1 + mx_1)}. \end{aligned}$$

The edges $du\xi_2 = \xi_1$ and $dl\xi_2 = \xi_1$ are the two sides of original triangle, hence we do not need to compute the other two linear functions as given in [6] as the solution to the optimization problem (15).

Rational function evaluated at the vertex (x_1, y_1) is $r(x_1, y_1) = 0/0$ where $\lim_{(x,y) \mapsto (x_1,y_1)} f(x, y) = f(x_1, y_1)$ while $r(x_2, y_2) = f(x_2, y_2)$ and $r(x_3, y_3) = f(x_3, y_3)$.

Verified symbolically in vertexNan.m

A.3.2 Optimization problems $(V - E^-)$ and $(V - E^+)$

The solutions obtained by solving the other two optimization problems, $(V - E^-)$ and $V - E^+$ are either infeasible or lie below the solution obtained by solving $V - E$. Thus the rational function obtained in (16) is the convex envelope over the entire triangle.

A.3.3 Example 2

The Convex Envelope of $f(x, y) = xy$ over $\text{conv}\{(1, 1), (0, 0), (2, 0)\}$ is the function $(x, y) \mapsto (2y^2)/(y - x + 2)$ on the same domain.

A.4 Two convex edges

Here we have two convex edge $y = mx + q$, hence the set E_P has two edges and the set V_P is empty. Using this information and referring to [6] we solve the following optimization problems in order to obtain the convex envelope.

$$\begin{aligned} \max\{\eta_i(a, b) + ax + by : \eta_i(a, b) &= \eta_j(a, b), \\ \eta_i(a, b) &\leq \eta_k(a, b), k \neq i, k \neq j, \\ (a, b) &\in S_r\}, \end{aligned}$$

where $V_1(x_1, x_2)$, $V_2(x_2, y_2)$ and $V_3(x_3, y_3)$ are the vertices of the triangle. Let (m_h, q_h) and (m_w, q_w) be the slopes and y-intercepts of the convex edges, V_1 is the vertex where these two edges intersect and let $x_2 = vx_1$ and $x_3 = vx_2$, then

$$x_1 = -\frac{q_h - q_w}{m_h - m_w}, y_1 = \frac{m_h q_w - m_w q_h}{m_h - m_w}, x_2 = vx_1, y_2 = q_h + m_h vx_1,$$

and $x_3 = vx_2, y_3 = q_w + m_w vx_2$.

A.4.1 Eta functions

Corresponding to the two convex edges, $(V_2 - V_1)$ and $(V_3 - V_1)$ we get the following η functions.

- For E_1 , $\eta_h = -(a + m_h b - q_h)^2 / (4m_h) - bq_h$
- For E_1^- , $\eta_h^- = f(x_2, y_2) - ax_2 - by_2 = x_2 y_2 - ax_2 - by_2$.
- For E_1^+ , $\eta_h^+ = f(x_1, y_1) - ax_1 - by_1 = x_1 y_1 - ax_1 - by_1$.
- For E_2 , $\eta_w = -(a + m_w b - q_w)^2 / (4m_w) - bq_w$
- For E_2^- , $\eta_w^- = f(x_3, y_3) - ax_3 - by_3 = x_3 y_3 - ax_3 - by_3$.
- For E_2^+ , $\eta_w^+ = f(x_1, y_1) - ax_1 - by_1 = x_1 y_1 - ax_1 - by_1$.

Thus we get nine optimization problems by taking all the pairs. Of these problems only one, $E_1 - E_2$ is feasible.

A.4.2 Optimization problem ($E_1 - E_2$)

In this problem, we consider the pair of η functions corresponding to the two convex edges. Thus

$$\eta_h = -\frac{(a + m_h b - q_h)^2}{4m_h} - bq_h \text{ and } \eta_w = -\frac{(a + m_w b - q_w)^2}{4m_w} - bq_w.$$

The optimization problem can now be written as

$$\max\{\eta_h + ax + by : \eta_w = \eta_h, (a, b) \in S_r\}.$$

Reformulating the constraint $\eta_h = \eta_w$, we get

$$\begin{aligned} -\frac{(a + m_h b - q_h)^2}{4m_h} - bq_h &= -\frac{(a + m_w b - q_w)^2}{4m_w} - bq_w, \\ \left(\frac{m_h - m_w}{4m_h m_w}\right)a^2 - 2\left(\frac{m_h q_w - m_w q_h}{4m_h m_w}\right)a + c &= 0, \end{aligned}$$

$$\text{where } c = \frac{-b^2 m_h^2 m_w + b^2 m_h m_w^2 - 2bm_h m_w q_h + 2bm_h m_w q_w + m_h q_w^2 - m_w q_h^2}{4m_h m_w}.$$

Solving for a , we get

$$a = \frac{m_h q_w - m_w q_h \pm \sqrt{m_h m_w}(q_h - q_w + bm_h - bm_w)}{m_h - m_w}.$$

Let $a = \alpha_1 b + \alpha_0$ with

$$\alpha_1 = \sqrt{m_h m_w},$$

$$\alpha_0 = \frac{m_h q_w - m_w q_h + \sqrt{m_h m_w}(q_h - q_w)}{m_h - m_w}.$$

The objective function is $\eta_h + ax + by$, by substituting η_h and a we get objective in terms of b . Similarly we get the bounds for b by substituting for a in $l_b \leq a + mb \leq u_b$. Thus we get the convex expression as

$$a_1x^2 + b_1xy + c_1y^2 + d_1x + e_1y + f_1$$

where

$$\begin{aligned} a_1 &= \frac{m_h m_w}{m_h + m_w + 2\sqrt{m_h m_w}}, \\ b_1 &= \frac{2\sqrt{m_h m_w}}{m_h + m_w + 2\sqrt{m_h m_w}}, \\ c_1 &= \frac{1}{m_h + m_w + 2\sqrt{m_h m_w}}, \\ d_1 &= \frac{(m_h q_w + m_w q_h)}{m_h + m_w + 2\sqrt{m_h m_w}}, \\ e_1 &= \frac{-(q_h + q_w)}{m_h + m_w + 2\sqrt{m_h m_w}}, \\ f_1 &= \frac{q_h q_w}{m_h + m_w + 2\sqrt{m_h m_w}}. \end{aligned}$$

or

$$\begin{aligned} a_1 &= \frac{m_h m_w}{m_h + m_w - 2\sqrt{m_h m_w}}, \\ b_1 &= \frac{-2\sqrt{m_h m_w}}{m_h + m_w - 2\sqrt{m_h m_w}}, \\ c_1 &= \frac{1}{m_h + m_w - 2\sqrt{m_h m_w}}, \\ d_1 &= \frac{(m_h q_w + m_w q_h)}{m_h + m_w - 2\sqrt{m_h m_w}}, \\ e_1 &= \frac{-(q_h + q_w)}{m_h + m_w - 2\sqrt{m_h m_w}}, \\ f_1 &= \frac{q_h q_w}{m_h + m_w - 2\sqrt{m_h m_w}}. \end{aligned}$$

By checking the bounds we get the domain as the entire triangle.

A.4.3 Example

Consider $f(x, y) = xy$ defined on the triangle with vertices, $V_1 = (2, 1)$, $V_2 = (0, 0)$, $V_3 = (1, 0)$. The vertex V_1 is the common vertex between the two convex edges. For the first convex edge $V_1 - V_3$, $m_h = 1$ and $q_h = -1$. For the second convex edge, $V_1 - V_2$, $m_w = \frac{1}{2}$ and $q_w = 0$.

The convex envelope has the same domain as f with expression

$$(x, y) \mapsto \frac{x^2 + 2\sqrt{2}xy + 2y^2 - x + 2y}{3 + 2\sqrt{2}}.$$

It is obtained from the second solution for a in Section A.4.2 by selecting the solution that is lower than f .

A.5 Three convex edges

In case we get a triangle with three convex edges, we can split it into two triangles with two convex edges each by adding a horizontal edge from the middle vertex.

B Conjugate of convex envelope of a bilinear function over a triangle

We improve the computational efficiency of [36] by symbolically computing formulae to obtain the conjugate corresponding to the convex envelopes in the above section. Following the steps given in [36], the domain is computed by computing the subdifferentials and the normal cones at each entity (vertices, edges and interior). As our functions are differentiable, the subdifferentials are the same as the gradients. In order to obtain the formulae we have MATLAB code that solves the required equations and gives us the required formulae. In this section, we have just stated the output (formulae) we have obtained which can be used to compute the required conjugates. Some of the code is given in the next section.

B.1 Triangle with no convex edges

For triangles with no convex edges, we have a linear function as the convex envelope as given in A.2. Thus the subdifferential (gradient) is a point, giving us a domain consisting of a normal cone from this point for each entity. For each edge as the normal cones are lines perpendicular to the edge, the subdifferential plus the normal cone is contained in a ray whereas for all the three vertices we get normal cones all starting from a common point(subdifferential). Thus the domain of the conjugate of a triangle with no convex edges is an unbounded polyhedral region. Now for each vertex $v = (v_x, v_y)$ to get the expression $f^*(s_1, s_2) = \max_v (s^T x - f(x)) = s_1 v_x + s_2 v_y - f(v_x, v_y)$.

B.2 Triangle with one convex edge

The convex envelope of a bilinear function on a triangle with one convex edge is a rational function. The formula is given by equation 16. At one vertex (x_1, y_1) , both the numerator and denominator evaluate to 0. Corresponding to this vertex we get the subdifferential as a parabola. The computation of the parabolic edge was done as shown in [36], using code given in C.2 and the formula obtained is given here as $as_1^2 + bs_1s_2 + cs_2^2 + ds_1 + es_2 + f$ where $a = -1$, $b = -2m$, $d = 2q + 4mx_1$,

$c = -m^2$, $e = -(2mq - 4my_1)$, and $f = -(q^2 + 4mx_1y_1)$. At the other two vertices we get polyhedral regions similar to the earlier section. For the edge opposite the vertex (x_1, y_1) , we get two parallel edges due to the normal cone bounded at the two other vertices giving us the parabolic region as explained in [36]. The conjugate expressions obtained on the regions corresponding to the three vertices is given by $f^*(s_1, s_2) = s_1v_x + s_2v_y - f(v_x, v_y)$, whereas the conjugate expression corresponding to the edge $y = mx + q$ is computed using the code given (C.1) below as $ax^2 + bxy + cy^2 + dx + ey + f$, where

$$a = \frac{1}{4m}, b = \frac{1}{2}, c = \frac{m}{4}, d = -\frac{q}{2m}, e = \frac{q}{2}, f = \frac{q^2}{4m}.$$

B.3 Triangle with two convex edges

Following the same approach as B.2, we obtain the conjugate domains as polyhedral and conjugate expressions corresponding to the vertices as $f^*(s_1, s_2) = s_1v_x + s_2v_y - f(v_x, v_y)$. Corresponding to the edges we get three linear edges, two of which are parallel. We get the parallel edges due to the normal cone at the edge and the third linear edge is due to subdifferential of the quadratic function obtained as a convex envelope for this type triangle. For the two convex edges we obtain the conjugate using the code and observe that it is the same expression as we obtained in B.2.

$$\frac{m_h^2 y^2 + 2m_h q_h y + 2m_h x y + q_h^2 - 2q_h x + x^2}{4m_h},$$

when the edge is $y = m_h x + q_h$.

C MATLAB code

C.1 Conjugate expression: code

Conjugate expression, $f^*(s) = \sup(s \cdot x - f(x))$ for f restricted to an edge.

```
function conj = conjugateExpr(edge,f,x,y)
    lambda = sym('lambda');
    infs = sym('inf');
    s1 = sym('s1');
    s2 = sym('s2');

    %edge
    dedge = sym.empty();
    dedge(1) = diff(edge,x);
    dedge(2) = diff(edge,y);

    %f
    df = sym.empty();
    df(1) = diff(f,x);
```

```

df(2) = diff(f,y);

eq1 = s1 - df(1) - lambda*dedge(1);
eq2 = s2 - df(2) - lambda*dedge(2);

xyl = solve([eq1,eq2,edge],[x,y,lambda]);

conj = infs;

if isempty(xyl)
    return
elseif isempty(xyl.x) | isempty(xyl.y) | isempty(xyl.lambda)
    return
end

conj = s1*xyl.x + s2*xyl.y - subs(f,[x,y],[xyl.x,xyl.y]);
conj = simplifyFraction(conj);
conj = subs(conj,[s1,s2],[x,y]);

end

```

C.2 Parabolic edge

Code to compute coefficients of the Parabolic edge.

```

t = sym('t');
% [x, y, const]
% psi2 = psi2(1)*x1+psi22*x2+psi20
% hence use index 3 for psi20 terms
% cpsi2 = obj.envelopeExpr(i).vpsi2.getLinearCoeffs (vars);
% cpsi1 = obj.envelopeExpr(i).vpsi1.getLinearCoeffs (vars);
% cpsi0 = obj.envelopeExpr(i).vpsi0.getLinearCoeffs (vars);
%
% vs1 = s1 - (2*cpsi1(1)*t - cpsi2(1)*t^2 + cpsi0(1));
% vs2 = s2 - (2*cpsi1(2)*t - cpsi2(2)*t^2 + cpsi0(2));
%
% vt = solve (cps2(2)*vs1 - cps2(1)*vs2, t);
% cps2(2)*vs1 - cps2(1)*vs2  cancels t^2 term - then solve
%   for t
% crs = subs(vs1,t, vt)      % crs = 0  equation of parabola

num = obj.envelope(i).f.getNum
den = obj.envelope(i).f.getDen
[r,q] = polynomialReduce(num,den)
r1 = factor(r)
r2 = r1(2)
if size(r1,2) == 3
    den = symbolicFunction((1/r1(1)) * den)
else
    den = symbolicFunction(den)
end
q = symbolicFunction(q)
r = symbolicFunction(r2)
cps2 = den.getLinearCoeffs (vars)
cps0 = q.getLinearCoeffs (vars)
cps1 = r.getLinearCoeffs (vars)

```

```

vs1 = s1 - (2*cpsi1(1)*t - cpsi2(1)*t^2 + cpsi0(1));
vs2 = s2 - (2*cpsi1(2)*t - cpsi2(2)*t^2 + cpsi0(2));

vt = solve (cpsi2(2)*vs1 - cpsi2(1)*vs2, t);
% cpsi2(2)*vs1 - cpsi2(1)*vs2 cancels t^2 term -
% then solve for t
crs = simplifyFraction(subs(vs1,t, vt))
% crs = 0 equation of parabola

end

```

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